

Aldric Labarthe

aldric.labarthe@ens-paris-saclay.fr | [Personal Page](#) | [in](#) [aldric-labarthe](#)

RESEARCH PROFILE

Researcher working at the intersection of geometry, statistical learning and network science, with a focus on developing mathematical and computational methods to understand complex systems and their underlying structure. Applications span social, economics and biomedical systems.

EDUCATION

- **University of Geneva & Ecole Normale Supérieure de Paris Saclay** 2025 - present
Joint Ph.D. Candidate in Mathematics & Computer Science Geneva, Switzerland & Gif-Sur-Yvette, France
 - **Ph.D. in Mathematics @ ENS Paris Saclay.** Supervised by Julien Randon-Furling (ENS).
 - **Ph.D. in Computer Science @ University of Geneva.** Supervised by Roland Bouffanais (UNIGE).
 - *Thesis Title: Geometry and Topology of Networks: Uncovering Structure through non-Euclidean Latent Manifolds.*
- **Ecole Normale Supérieure de Paris Saclay** 2022 - 2025
Normalien Student (Élève Normalien) – Bachelor, Master's degree Gif-Sur-Yvette, France
 - 2024 - 2025: **Master's degree** *Mathématiques, Vision, Apprentissage (MVA)* in applied mathematics and AI
 - 2023 - 2024: **Master's degree** *Master of Economics* in quantitative economics and mathematics.
 - 2022 - 2023: **Bachelor's degree**, economics major and electives in finance, mathematics and management.
- **Lycée Turgot & Université Panthéon-Sorbonne** 2020 - 2022
Preparatory class Paris, France
 - Intensive 2-year class in Mathematics and Economics with first class honours.
 - **Result:** admitted into the Ecole Normale Supérieure de Paris Saclay, as **major in the competitive exam.**

WORK EXPERIENCE

- **École nationale de la statistique et de l'administration économique (ENSAE) ** September 2025 - present
Graduate Program Coordinator, Department of Statistics and Machine Learning Palaiseau, France
 - Recruited, onboarded, and managed 25+ Teaching Assistants across statistics and ML programs.
 - Designed and updated graduate curricula in statistical learning and optimization; delivered 65+ hours of advanced graduate lectures.
- **Centre Borelli (ENS Paris-Saclay) & Applied Complexity Group (University of Geneva)** April 2025 – September 2025
Graduate Machine Learning Researcher Paris & Geneva
 - Developed topology-aware graph representation algorithms, improving structure-preserving node embeddings on empirical networks.
 - Engineered C++ and Python models analyzing latent geometric manifolds on attributed graphs, co-authoring research during a 4-month research residency in Geneva.
- **BeyondSolutions ** May 2023 - September 2023
Financial Analyst Intern Paris, France
 - Architected end-to-end econometric pipelines on 10-year financial and corporate datasets using time-series linear modeling.

TRANSLATIONAL AI & R&D INITIATIVES

- **Whole-body PET-CT AI Platform (Glync.ai)** November 2024 - present
Collaboration with Hôpitaux Universitaires Paris-Saclay AP-HP & Orsay statistics lab (UMR 8628) 
 - Co-authored two papers on whole-body PET-based AI biomarkers
 - Patent pending on probabilistic organ interaction modeling (submission in progress).
 - Secured data partnership with AP-HP, unlocking the largest European cohort for model validation on PAH disease.
 - Winner of the “Prix Audace” @ Starthèse Paris-Saclay 2026: awarded a fully funded Learning Expedition & pitch residency at MIT.

CONFERENCES AND WORKSHOPS

- **CompleNet 2026** (Long presentation) Universidad de Zaragoza, Zaragoza (ES) May 2026
- **CREST Seminar** (Long presentation) Institut Polytechnique de Paris, Palaiseau (FR) February 2026
- **CEPS Workshop** (Long presentation) Ecole Normale supérieure Paris-Saclay, Gif-Sur-Yvette (FR) November 2025
- **SiDLab Workshop 2025** (Short presentation) Global Studies Institute, Geneva (CH) June 2025

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, B=PATENT, S=IN SUBMISSION, T=THESIS, P=PREPRINT

- [S.4] A. Labarthe, F. Besson, et al. (2026). **Geometric stabilization reveals the limits of perturbation-based individualized connectomics**. Manuscript submitted for publication.
- [S.3] A. Labarthe (2026). **Bayesian Latent Space Models for Graphs Are Misspecified: Toward Robust Inference via Generalized Posteriors**. Manuscript under review.
- [S.2] A. Labarthe, R. Bouffanais, J. Randon-Furling (2026). **Aligning the Unseen in Attributed Graphs: Interplay between Graph Geometry and Node Attributes Manifold**. Manuscript submitted for publication.
ArXiv preprint: *arXiv:2601.22806*.
- [S.1] A. Labarthe, Y. Kerzreho (2025). **From Technical Feasibility to Substitutability: A Geometric Theory of Differentiation**. Manuscript under review.
ArXiv preprint: *arXiv:2507.01985*.
- [J.2] A. Labarthe, F. Besson, et al. (2026). **Personalized mapping of body homeostasis using whole-body PET connectomics and routine FDG PET imaging**. *Communications Medicine*, DOI: 10.1038/s43856-026-01549-y
- [J.1] A. Labarthe, Y. Kerzreho (2026). **Generating social networks with static and dynamic utility-maximization approaches**. *Social Network Analysis and Mining*, Vol. 16, Issue 51 DOI: 10.1007/s13278-026-01582-8
- [T.1] A. Labarthe (2024). **Strategies and equilibria on selected markets: a multi-agent simulation and stochastic modeling approach**. Master thesis advised by J. Randon-Furling, graded 19/20 by the ENS Paris-Saclay jury (best thesis). *Currently working on a manuscript for a Physics journal*.

TEACHING EXPERIENCE (FULL LECTURE + TUTORIALS)

- **Mathematics and Statistical Learning for Economists** 2026
Graduate level (M2) ENSAE Paris
- **Introduction to Statistical Learning** 2026
Graduate level (M1) Ecole Polytechnique

TEACHING EXPERIENCE (TUTORIALS)

- **AI: Applied Statistical Learning** 2025-2027
Graduate level (M2), ENSAE Paris Teaching Assistant of A. Dalalyan
- **AI: Optimal Transport** 2025-2027
Graduate level (M2), ENSAE Paris Teaching Assistant of M. Cuturi
- **Computer Science: Python for Data Science** 2025-2027
Graduate level (M1), ENSAE Paris Teaching Assistant of L. Galiana
- **Mathematics: Information theory** 2025-2026
Graduate level (M2), ENSAE Paris Teaching Assistant of A. Khaleghi
- **Mathematics: Statistics** 2025-2026
Graduate level (M1), ENSAE Paris Teaching Assistant of A. Dalalyan
- **Mathematics: Multivariate analysis and metric spaces topology** 2025
Second year, Bachelor of mathematics (Université Panthéon-Sorbonne) Teaching Assistant of G. Groce
- **Microeconomics: Uncertainty, insurance and game theory** 2024-2025
Third year, Bachelor of economics (Université Panthéon-Sorbonne) Teaching Assistant of F. Bloch
- **Microeconomics: Decision theory and consumer theory** 2024-2025
First year, Bachelor of mathematics (Université Panthéon-Sorbonne) Teaching Assistant of S. Gauthier

CERTIFICATIONS

- **International English Language Testing System (IELTS)** Score: 7.5 (CEFR: C1+) november 2023
- **TOEIC**: Score: 940 / 990 (CEFR: C1) march 2022

ADDITIONAL INFORMATION

Languages: French (Proficiency level), English (Proficiency level)

Interests: Hiking, climbing (indoor or outside, bouldering and lead climbing)